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跳躍擴散模型下離散型回顧選擇權之評價

Pricing Discrete Lookback Options

Under A Jump-Diffusion Model

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一直以來，抱持著盡人力聽天命的態度去接受眼前的挑戰，一關又一關總是得到很多人的幫助而成功，研究所這兩年更是讓我從層層挑戰當中躍進許多。這首先我最要感謝傅承德老師的指導，讓我體驗了一段很充實的研究旅程；除此之外，老師教給我的，是努力不懈的學習態度和充滿熱誠的研究精神，讓我充滿動力去接受未來的挑戰。

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摘要

所謂回顧選擇權乃是一項契約明定投資者的報酬受到標的資產極值的影響,此種選擇權能保證給予投資者最大的報酬。大多數模型在評價回顧選擇權時,都假設連續時間觀測標的資產價格之極值,並且都能推導出精確的封閉解。然而在實務上,許多有回顧型性質的契約,都會明定特定的(離散)觀測時點,此類的選擇權即稱為離散型回顧選擇權。在本篇研究中,我們理論推倒出了在固定跳躍擴散模型(constant jump diffusion model)之下,利用修正過後之連續型回顧選擇權公式來逼近離散型回顧選擇權的價格。使用的方法參考 Broadie et al. (1997),這篇論文曾經指出,在幾何布朗運動架構之下,離散型回顧選擇權價格可以用修正後之連續型回顧選擇權公式來逼近。當中我們應用了 Siegmund (1985) 中的技巧來得到我們的結果。由論文最後數值分析之結果可以看出此逼近公式之高度正確性。

關鍵字: 回顧選擇權, 跳躍擴散模型, 調整擴散逼近, Laplace 轉換.

Abstract

A lookback option introduced in 1979 by Goldman et al. is a path dependent option settles based upon the maximum or minimum of the underlying price process achieved during the entire life of the option. Most models for pricing lookback options assume continuous monitoring of the extreme and have closed solutions. However, in practice, many real contracts with lookback provisions specify discrete monitoring times. Such options are called discrete lookback options. In this article, we focus on pricing discrete lookback options using continuous lookback formulas by applying a simple continuity correction under the constant jump diffusion model. We use the same method of correction as Broadie et al. (1999) which have solved such problems under the geometric Brownian motion setting. The correction is justified theoretically by applying the techniques from sequential analysis, particularly Siegmund (1985). And we also give numerical results.

Keywords: lookback options, jump diffusion model, continuity correction, Laplace transform.

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1 Introduction

A *lookback option* introduced in 1979 by Goldman et al. is a path dependent option settles based upon the maximum or minimum of the underlying price process achieved during the entire life of the option. Essentially, at expiration, the holder can ‘look back’ over the entire contract period and exercise based upon the optimal value of the underlying asset achieved during that period. Lookbacks can be structured as puts or calls and come in two basic forms: *fixed strike and floating strike*.

A floating strike lookback call gives the holder the right to buy the underlying asset at the lowest realized price that was observed during the options lifetime while a floating strike lookback put allows the holder to sell at the highest realized price over the period. That is, if $S(t)$ denotes price process of the underlying asset at time t in the lifetime $[0, T]$ of these options, then the terminal payoffs of the European floating calls and European floating puts can be respectively expressed as

$$S(T) - \min_{t \in [0, T]} S(t) \quad \text{and} \quad \max_{t \in [0, T]} S(t) - S(T).$$

In contrast, a fixed strike lookback call (resp. put) gives the holder the right to buy (sell) the maximum (minimum) underlying asset price achieved during the lifetime for the options by using the fixed strike price K . And the time- T payoffs of the European fixed strike lookback calls and puts with strike K can be expressed as

$$\max \left\{ \max_{t \in [0, T]} S(t) - K, 0 \right\} \quad \text{and} \quad \max \left\{ K - \min_{t \in [0, T]} S(t), 0 \right\}.$$

One attraction for the speculative investor of exotic options in general is the opportunity to obtain greater leverage than would normally be achieved by using plain vanilla options. In particular, options with a lookback feature may also be used to hedge against large fluctuations in the price of the underlying security, or to solve the dilemma about the optimal timing of entry into (or exit from) the market. Moreover, a charming benefit of lookback options is that the options would guarantee the investor's fantasy of "buy at the low and sell at the high". It means that they are never out-of-the-money, but the result is that lookbacks are often more expensive than similar vanilla style options.

Under the Black-Scholes option pricing framework, explicit valuation formulas have been derived for European lookback options. Goldman, Sosin and Gatto (1979) and Conze and Viswanathan (1991) followed a probabilistic approach in taking the discounted expectation of the terminal payoffs. By making use of the PDE framework, Xu and Kwok (2005) solved the Black-Scholes partial differential equations to obtain the option value. Buchen and Konstandatos (2005) demonstrated that a lookback option can be considered as an integrated form of a related barrier option.¹

¹The relationship between barrier options and lookback options seem quite close from this paper. But the issues have received little attention in the literature and we will give a related discussion on the issues later.

Otherwise, Various authors have derived formulae for the valuation of lookback options under Lévy processes model: Kou and Wang (2003, 2004) derived explicit formulas for the values of lookback options in a jump diffusion model where the jumps are double-exponentially distributed; they make use of a special property of the exponential distribution, namely the memoryless property, which allows them to explicitly calculate the overshoot distribution. By making use of fluctuation theory, Lipton (2002) derived similar formulas under the same model. Recently, Nguyen-Ngoc (2003) applied probabilistic results in fluctuation theory for Lévy processes and derived formulae in general exponential Lévy models, which can be numerically evaluated with the use of Fourier inversion algorithms in two and three dimensions.

One important and interesting feature of lookback option is that the values are quite sensitive to whether the extreme are monitored discretely or continuously. The formulae in the articles cited above are appropriate when the extreme of the price process is continuously monitored. However, in practice, many contracts with lookback features are settled by reference to a discrete sampling of the price process at regular time intervals (for example, daily at 10:00 am). These options are usually referred to as *discrete* lookback options. In these circumstances the continuous-sampling formulae are inaccurate.

Although discretely monitored lookback options are popular and important, pricing them is harder. In contrast with continuously monitored lookback options, the

price of a discrete lookback option does not have a closed-form pricing formula in general. Moreover, the exact pricing results available for continuous lookback options do not extend to the discrete case. Numerical methods using standard lattice techniques or Monte Carlo simulation face significant difficulties in incorporating discrete monitoring, see, for example, Boyle et al. (1997) and Broadie et al. (1999), which indicate that there can be substantial price differences between the discrete lookback and its continuous counterpart. Even for daily sampling the difference is significant.

Discretely monitored options have received much less attention in the literature than their continuous time counterparts. Broadie et al. (1997) provided a technique which has given remarkably good results for approximation pricing discrete barrier options. They used Siegmund's corrected diffusion approximation, referred to Siegmund (1985) and gave a simple continuity correction for relating the prices of discrete- and continuous-time versions of barrier options. Later, they derived the price of discrete lookback options in Broadie et al. (1999). Although these two approaches are same, the techniques are different.

The results of Broadie et al. are all derived under Black-Scholes model. However, it is known that much research has been conducted to modify the Black-Scholes model based on Brownian motion and normal distribution in order to incorporate two empirical features: the asymmetric leptokurtic and the volatility smile. The

former means that the asset return distribution is skewed to the left, and has a higher peak and two heavier tails than those of the normal distribution. And the latter indicates that, if the Black-Scholes model is correct, then the implied volatility should be constant, while it is widely recognized that the implied volatility curve resembles a smile, meaning that it is a convex curve of the strike price.

One of many models capturing these features is so-called jump diffusion model which is a particular case of Lévy processes. And it also captures the discontinuities of many financial processes showed by many empirical studies. Discontinuity (jump) is usually regarded as the result when a big event (rare event) occurred, such as bankruptcy of a company or market crash. Furthermore, for an argument in favor of using Lévy processes in financial modeling, we refer to Leblanc and Yor (1998). That motivates us to study pricing discrete lookback options under a jump-diffusion model. There are many kinds of jump diffusion model, and the simplest one is constant jump diffusion model. That is, the jump size is assumed to be constant and the jump frequency is described by a pure Poisson process. Though this model is simple, its parameters have a nice economic interpretation and it is easy to calibrate.

Due to the jump part, the market is incomplete and so it is impossible to construct a hedging portfolio with stocks and riskless assets. Contrary to the case of a complete market in which only one equivalent martingale measure exists, there are infinite numbers of equivalent martingale measures in incomplete market. How-

ever, we still can identify *the only* market pricing measure, that is, the equivalent martingale measure (risk-neutral probability measure) to be used to price derivative products. As for how to change the measure from original (objective) probability to a risk-neutral probability measure, see, for example, Shreve (2004); and for how to choose the market pricing measure among risk-neutral probabilities, we refer to Brockhaus et al. (2000) which is focusing on risk minimizing strategy and its associated minimal martingale measure under the jump-diffusion processes.

There are numerical approaches for pricing such an option under the framework with jumps, see, for example, Petrella and Kou (2004) which uses Laplace transform method. However, this work focuses on the connection between the prices of discrete and continuous lookback options just like those in Broadie et al. (1997), and Kou (2003). The main idea follows the framework of Kou (2003) and uses the results of Siegmund's corrected diffusion approximation. Kou, Petrella and Wang (2005) also used Laplace transforms method. They inverted Laplace transform via the Euler inversion, which does not require high-precision calculation and lead to fast and accurate results of lookback options.

The rest of this thesis is organized as follows. The mathematical formulation of the problem is given in Section 2, in which we also describe how we price a continuous or a discrete lookback option and presents corrections for continuous-time formulas to price discrete-time options approximately, i.e Theorem 2.1 and 2.2, which are our

main results. In Section 3, we give the proofs of the main results, while Section 4 develops numerical methods for pricing discrete-time and continuous-time options accurately. Conclusion remarks and Further research are given in the last section.

2 Pricing Lookback Options

2.1 Continuous-monitoring case

Let $(\Omega, \mathcal{F}, P)$ be a probability space, to which we define a Brownian motion $\overline{W}(t)$, $0 \leq t \leq T$, and a Poisson process $N(t)$ with intensity $\overline{\lambda}$. Assume $\overline{W}(t)$ and $N(t)$ are independent. Let $\mathcal{F}(t)$, $0 \leq t \leq T$, be the filtration generated by the Brownian motion and the Poisson process. Set

$$\overline{M}(t) = N(t) - \overline{\lambda}t.$$

Then $\overline{M}(t)$ is the compensated Poisson process under the actual probability measure P . In addition, we denote the martingale probability measure P^* , which is “the” risk-neutral probability measure.² Then the P^* -dynamics of the underlying asset price, $S(t)$, will be modeled by the stochastic differential equation (SDE)

$$\begin{cases} dS(t) = rS(t)dt + \sigma S(t)dW(t) + (\delta - 1)S(t-)dM(t), \\ S(0) = s, \end{cases} \quad (1)$$

where r is the risk free rate, σ is the volatility and $(\delta - 1)$ is the relative jump size, respectively. Moreover, $W(t)$ is a P^* -Wiener process and $M(t) = N(t) - \lambda t$ is a compensated Poisson process under P^* with intensity λ .

Note that here λ is the average number of jumps within a year and the price $S(t) = \delta S(t-)$ if t is a jump point and assume that $\delta > 0$ in order for the price

²In order to price the derivatives, we have to suppose that the market is complete. In other words, to decide a unique martingale measure is necessary. Moreover, the present value of future financial flows equals their expectation under P^* after discounting, i.e. $e^{-rt}S_t$ is a martingale.

to remain positive. The jump size is strictly negative if $0 < \delta < 1$ and the asset price will have a large slump when a negative jump occurs. The model under this assumption is called *spectrally negative jump processes*. These are real valued random processes with stationary independent increments which have no positive jumps. On the other hand, the jump size is strictly positive if $\delta > 1$, in which the asset price will have a large increase at an appearance of positive jumps, and the model is called *spectrally positive jump processes* which have no negative jumps.

$S(t)$ defined in (1) can be written explicitly as

$$\begin{aligned} S(t) &= s \exp\left\{\left(r - \frac{1}{2}\sigma^2\right)t + \sigma W(t) - \lambda(\delta - 1)t\right\} \delta^{N(t)} \\ &= s \exp\{(\mu + \lambda \log \delta)t + \sigma W(t) + M(t) \log \delta\} \\ &\equiv s \exp\{X(t)\}^3, \quad t \in [0, T], \end{aligned}$$

where $\mu = r - \frac{1}{2}\sigma^2 - \lambda(\delta - 1)$. From the risk neutral valuation theorem⁴, the price of a continuous lookback put (LBP from now on) option at an arbitrary time $0 < t < T$ is given by

$$V = e^{-r(T-t)} \mathbb{E}^* \left[\max_{t \in [0, T]} S(t) - S(T) \middle| \mathcal{F}_t \right].$$

Obviously, the arbitrage free price is given as the discounted value (with the discount factor being $e^{-r(T-t)}$) of the mathematical expectation, taken with respect to the risk-neutral probability P^* , of the future claim.

³ $X(t)$ can be regarded as a special case of Lévy processes with the triple $[\mu + \lambda \log \delta, \sigma, \lambda \delta(1)]$.

⁴Refer to Björk (2004), chapter 7. The main idea is that the value of the contingent claim can be calculated as if we live in a risk neutral world with the money market account as the numeraire.

Denote $S_+(t) = \max_{u \in [0, t]} S(u)$ and rewrite $\max_{t \in [0, T]} S(t)$ as $\max[S_+(t), \max_{u \in [t, T]} S(u)]$.

In a similar way, the price of a continuous lookback call (LBC from now on) depends on $S_-(t) = \min_{u \in [0, t]} S(u)$ and $\min_{u \in [t, T]} S(u)$. We refer to $S_-(t)$ and $S_+(t)$ as the *predetermined min* and *max*, respectively. To simplify the notation, we use symbols S_+ and S_- , which omit the (t) , in the sequel. Besides, let $M = \max_{u \in [t, T]} X(u)$ denote the maximal $X(u)$ after a given time t , then we get $\max_{u \in [t, T]} S(u) = se^M$. Therefore, the price of LBP becomes

$$\begin{aligned} V(S_+) &= e^{-r(T-t)} \mathbb{E}^*[S_+ + (\max_{u \in [t, T]} S(u) - S_+)^+ - S(T) | \mathcal{F}_t] \\ &= e^{-r(T-t)} S_+ + e^{-r(T-t)} \mathbb{E}^*[(se^M - S_+)^+ | \mathcal{F}_t] - S(t), \end{aligned} \quad (2)$$

in which the second equation holds because $e^{-r(T-t)} \mathbb{E}^*[S(T) | \mathcal{F}_t] = S(t)$. For the key-term $\mathbb{E}^*[(se^M - S_+)^+ | \mathcal{F}_t]$, I calculated in detail as follows: let $\tau(c, Y) := \inf\{t > 0 : Y(t) \geq c\}$ and denote $a = \log(\frac{S_+}{s})$, a constant.

$$\begin{aligned} &\mathbb{E}^*[(se^M - S_+)^+ | \mathcal{F}_t] \\ &= \int_{-\infty}^{\infty} (se^x - S_+)^+ dP^*(M \leq x) \\ &= \int_a^{\infty} se^x P^*(M \geq x) dx \\ &= \int_a^{\infty} se^x P^*(\tau(x, X) \leq T) dx. \end{aligned}$$

In above equations, the first equation holds because the expectation is just an integral with respect to the density in x of the maximal distribution $P^*(M \geq x)$. Integration by parts yields the second one, and the last equation uses the fact that

for $b \in R^+$,

$$\{se^M \geq b\} = \{\tau(b, se^X) \leq T\}^5.$$

The price of LBP thus becomes

$$V(S_+) = e^{-r(T-t)}S_+ - S(t) + e^{-r(T-t)} \int_a^\infty se^x P^*(\tau(x, X) \leq T) dx. \quad (3)$$

Remark 1. Focus on $e^{-r(T-t)}E^*[(se^M - S_+)^+ | \mathcal{F}_t]$ which can be deemed the discounted value of a Up-and-In call option with barrier S_+ and strike price S_+ , called the *moving barrier option* since S_+ is a moving constant depending on the process of the asset price during $[0, t]$. The difference between them is that the underlying asset in barrier option should be replaced by its maximum observed in $[t, T]$. On the other hand, it can also be regarded as the European fixed strike lookback call options with strike price S_+ . Therefore, we can find the relation between the fixed strike LBC and the floating strike LBP. Moreover, to get the value of fixed strike LBC options under a jump diffusion model. These issues are quite interesting and will be included in our further researches. The latter will be open for later discussion.

⁵From the fact we can transform the troublesome extreme problem into stopping time issues.

2.2 Discrete-monitoring case

Consider the extreme monitored only at time $i\Delta t$, $i = 1, 2, \dots, m$, where $\Delta t = \frac{T}{m}$, $T > 0$ is fixed, and m is the frequency of monitoring. Then under the risk neutral probability P^* , the discrete version of our asset price, S_n , is given by

$$\begin{aligned} S_n &= s \exp \left\{ \mu n \Delta t + \sigma \sqrt{\Delta t} \sum_{i=1}^n Z_i + \log \delta \sum_{i=1}^n N_i \right\} \\ &= s \exp \left\{ \sum_{i=1}^n [\sigma \sqrt{\Delta t} Z_i + (\mu + \lambda \log \delta) \Delta t + M_i \log \delta] \right\} \\ &\equiv s \exp \{X_n\}, \quad n = 1, 2, \dots, m, \end{aligned}$$

where $Z_i \stackrel{i.i.d}{\sim} N(0, 1)$, $N_i \stackrel{i.i.d}{\sim} \text{Poisson}(\lambda \Delta t)$, and $M_i = N_i - \lambda \Delta t$. Also note that $\{Z_i\}$ and $\{N_i\}$ are independent and $X_0 \equiv 0$.

By analogy, we can also apply the risk-neutral valuation theorem to get the price of a discrete LBP option at the k^{th} fixed date, which is given by

$$V_m = e^{-r(m-k)\Delta t} \mathbb{E}^*[\max_{0 \leq n \leq m} S_n - S_m | \mathcal{F}_k],$$

where $\mathcal{F}_k$ is the natural filtration of discrete underlying asset prices. Similarly, we rewrite $\max_{0 \leq n \leq m} S_n$ as $\max\{S_+, \max_{k \leq n \leq m} S_n\}$, where $S_+ = \max_{0 \leq n \leq k} S_n$. Define notations $M_m = \max_{k \leq n \leq m} X_n$ as the maximal X_n after the given time $k\Delta t$ and $\tau'(c, Y) := \inf\{n \geq 1 : Y_n \geq c\}$. Then, the price of discrete LBP would become

$$\begin{aligned} V_m(S_+) &= e^{-r(m-k)\Delta t} \mathbb{E}^*[S_+ + (\max_{k \leq n \leq m} S_n - S_+)^+ - S_m | \mathcal{F}_k] \\ &= e^{-r(m-k)\Delta t} S_+ + e^{-r(m-k)\Delta t} \mathbb{E}^*[(se^{M_m} - S_+)^+ | \mathcal{F}_k] - S_k, \end{aligned} \quad (4)$$

where $E^*[(se^{M_m} - S_+)^+ | \mathcal{F}_k]$ can be calculated in detail as follows,

$$\begin{aligned}
& E^*[(se^{M_m} - S_+)^+ | \mathcal{F}_k] \\
&= \int_{-\infty}^{\infty} (se^x - S_+)^+ dP^*(M_m \leq x) \\
&= \int_a^{\infty} se^x P^*(M_m \geq x) dx \\
&= \int_a^{\infty} se^x P^*(\tau'(x, X) \leq m) dx,
\end{aligned}$$

in which the last equation holds since the fact that for $b \in R^+$,

$$\{se^{M_m} \geq b\} = \{\tau'(b, se^X)\Delta t \leq T\}.$$

Then the price of discrete LBP is given by

$$V_m(S_+) = e^{-r(m-k)\Delta t} S_+ - S_k + e^{-r(m-k)\Delta t} \int_a^{\infty} se^x P^*(\tau'(x, X) \leq m) dx. \quad (5)$$

Note that the $S_{\pm}$, in continuous as well as in discrete version, will be assumed as the same value when they are calculated for discrete and continuous options. And when an option is valued, the corresponding $S_{\pm}$ is known and acts as a parameter of the payoff function, just like the strike price in a standard option.

2.3 Continuity correction

This section focuses on continuity correction for relating the prices of discrete- and continuous-time versions of lookback options. The reason is that the analytic solutions of continuous lookback options are not difficult to derive. For example, under the Black-Scholes model the value of continuous floating strike LBP options is given by

$$S_+ e^{-rT} N(-d_1 + \sigma\sqrt{T}) - s N(-d_1) \\ + e^{-rT} \frac{s\sigma^2}{2r} \left[e^{-rT} N(d_1) - \left(\frac{s}{S_+}\right)^{-2r/\sigma} N\left(d_1 - \frac{2r}{\sigma}\sqrt{T}\right) \right],^6$$

where

$$d_1 = \left(\log \frac{s}{S_+} + rT + \frac{1}{2}\sigma^2 T \right) / \sigma\sqrt{T}.$$

Moreover, under a jump diffusion model, Kou and Wang (2004) have also derived explicit formulae in terms of Laplace transforms for the values of lookback options. Numerical examples indicate that the formulae are easy to implement, and are accurate for continuous time option price.

We observe that the values of continuous lookback options are easy to obtain. Discrete lookback options, however, are more practical in our trading market. For this reason, we would like to find the connection between the prices of discrete and continuous lookback options. And then, to price discrete lookback options.

⁶The formula derived by Goldman et al.(1979) can be seen as the sum of the value of a standard option given by the Black-Scholes formula plus another term. And it is consistent with our intuition that lookbacks are often more expensive than similar vanilla style options

The following theorem is our main result:

Theorem 2.1. *Let V be the price of a continuous floating strike lookback put or call, and V_m be the price of an otherwise identical floating strike discrete lookback. Then under the spectrally negative jump diffusion model, i.e. $0 < \delta < 1$, the price of a discrete lookback at the k^{th} fixing date and the price of a continuous lookback at time $t = k\Delta t$ satisfy*

$$V_m(S_+) = \left[e^{-\beta C \sqrt{\Delta t}} V(S_+ e^{\beta C \sqrt{\Delta t}}) + (e^{-\beta C \sqrt{\Delta t}} - 1) S_t \right] + o\left(\frac{1}{\sqrt{m}}\right) \quad (6)$$

for puts option.

Under the spectrally positive jump diffusion model, i.e. $\delta > 1$, we have an approximation

$$V_m(S_-) = - \left[e^{\beta C \sqrt{\Delta t}} V(S_- e^{-\beta C \sqrt{\Delta t}}) + (e^{\beta C \sqrt{\Delta t}} - 1) S_t \right] + o\left(\frac{1}{\sqrt{m}}\right) \quad (7)$$

for calls option. Here the constants $C = \sqrt{\sigma^2 + \lambda(\log \delta)^2}$ and $\beta = -\zeta(\frac{1}{2})/\sqrt{2\pi} \approx 0.5826$ with ζ being the Riemann zeta function.

We can interpret this result as saying that to value a discrete lookback put using the continuous time formula, to which one should first inflate the predetermined max by a factor of $e^{\beta C \sqrt{\Delta t}}$, then deflate the expected maximum over $[0, T]$ by the same factor. Similarly, for a lookback call, one first deflates the predetermined min, then inflates the expected minimum.

Due to the jump part, when a jump diffusion process crosses the boundary level ⁷, sometimes it may incur an ‘overshoot’ over the boundary. In general, the distribution of the overshoot is not known analytically except when the jumps are double-exponentially distributed. To simplify the troublesome problems and to focus on the corrected diffusion approximation issues, in this paper, we consider the class of spectrally negative ($0 < \delta < 1$) jump processes for LBP while spectrally positive ($\delta > 1$) jump processes for LBC. In this case, the overshoot phenomenon only happens with discretely monitoring case, it means that the overshoot is only due to discretization effect like the assumption in Broadie et al. (1997). Therefore our formulae generalized that in Broadie et al. (1999) and coincide when $\delta = 1$.

Various authors have treated such problems in the case where the driving process is a spectrally positive or negative Lévy process; see, for example, Avram et al. (2004) in which they consider the case of spectrally negative jump diffusion whereby the underlying asset exhibits no positive jumps. Such processes are often considered in the context of the theories of dams, queues, insurance risk and continuous branching processes. The reader is referred to Prabhu (1981) for details.

As for the fixed strike lookback options, thanks to Remark 1, and the value processes of the floating strike lookback options derived in section 2, we have the relation between the floating strike LBP and the fixed strike LBC at an arbitrary time $t \in [0, T]$. Let $F^c(S_+)$ denote the continuous fixed strike LBC with strike

⁷In this paper, we can regard S_+ as the boundary for LBP during $[t, T]$.

price $K = S_+$ and $V^p(S_+)$ denote the continuous floating strike LBP with the predetermined $\max S_+$, then from the definition of time-T payoffs of the European fixed strike lookback calls we have

$$\begin{aligned}
F^c(S_+) &= e^{-r(T-t)} \mathbb{E}^* \left[(\max_{t \in [0, T]} S(t) - S_+)^+ | \mathcal{F}_t \right] \\
&= e^{-r(T-t)} \mathbb{E}^* \left[(\max\{S_+, \max_{u \in [t, T]} S_u\} - S_+)^+ | \mathcal{F}_t \right] \\
&= e^{-r(T-t)} \mathbb{E}^* \left[(\max_{u \in [t, T]} S(u) - S_+)^+ | \mathcal{F}_t \right] \\
&= V^p(S_+) - e^{-r(T-t)} S_+ + S_t,
\end{aligned} \tag{8}$$

in which the last equation holds for the equation (2). Similarly, from the equation (4), the discrete versions of these options satisfy

$$F_m^c(S_+) = V_m^p(S_+) - e^{-r(m-k)\Delta t} S_+ + S_k, \tag{9}$$

at $k = 1, \dots, m$, where $F_m^c(S_+)$ denote the discrete fixed strike LBC with strike price $K = S_+$ and V_m^p denote the discrete floating strike LBP options. The equations (8) and (9) are useful since they show that any numerical methods for pricing floating strike lookback options can be used to price fixed strike lookbacks.

Then, we can get the continuity correction for fixed strike LBC case. That is presented as follows.

Theorem 2.2. *Let F be the price of a continuous fixed strike lookback put or call with strike price S_+ , and F_m be the price of an otherwise identical fixed strike discrete lookback. Then under the spectrally negative jump diffusion model, i.e. $0 < \delta < 1$, the price of a discrete lookback at the k^{th} fixing date and the price of a continuous lookback at time $t = k\Delta t$ satisfy*

$$F_m(S_+) = e^{-\beta C \sqrt{\Delta t}} F(S_+ e^{\beta C \sqrt{\Delta t}}) + o\left(\frac{1}{\sqrt{m}}\right) \quad (10)$$

for calls option.

Under the spectrally positive jump diffusion model, i.e. $\delta > 1$, we have the approximation

$$F_m(S_-) = e^{\beta C \sqrt{\Delta t}} F(S_- e^{-\beta C \sqrt{\Delta t}}) + o\left(\frac{1}{\sqrt{m}}\right) \quad (11)$$

for puts option. Here the constants $C = \sqrt{\sigma^2 + \lambda(\log \delta)^2}$ and $\beta = -\zeta(\frac{1}{2})/\sqrt{2\pi} \approx 0.5826$ with ζ being the Riemann zeta function.

Note that the fixed strike LBP options can be related to floating strike LBC options, and the same procedure can be used to derive the option price formulae. These formulae also generalized that in Broadie et al. (1999) and coincide when $\delta = 1$, where fixed strike lookbacks are called *hindsight options*.

3 Proofs of the Main Results

In this section we merely directly prove equation (6) in detail for LBP options, that is under the case $0 < \delta < 1$. Since the proof for LBC case is similar, we will not repeat it here. We first observe that the only difference between the values of continuous-(3) and discrete-monitoring(5) LBP options is their integrands. More precisely, the distributions of $\tau(x, X)$ and $\tau'(x, X)\Delta t$ are consequently attention-getting to accomplish this problem. Therefore, in order to connect the two prices, we need to study the relation between the distributions of $\tau(x, X)$ and $\tau'(x, X)\Delta t$ first. The following results are needed in our proofs.

3.1 Distribution of the first passage times

Let $\psi_t(\theta) = \log E^*[e^{\theta X(t)}]$ be the cumulative generating function of the jump processes $X(t)$. It is easy to see that they are given by

$$\begin{aligned}\psi_t(\theta) &= t \cdot \left\{ \mu\theta + \frac{1}{2}\sigma^2\theta^2 + \lambda(e^{\theta \log \delta} - 1) \right\} \\ &\equiv t \cdot G(\theta),\end{aligned}$$

where $G(\theta) := G_1(\theta)$ is called the *characteristic exponent* of $X(1)$. Besides, one can show that $G(\theta)$ is a nonnegative, convex function of θ with $\theta = \theta_1 < 0$ such that $G'(\theta_1) = 0$. It means that G is one-to-one for θ greater than θ_1 .

Moreover, the Laplace transform of the continuous first passage times, which is convenient for numerical Laplace inversion is as follows:

Lemma 3.1. *The Laplace transform of τ is given by*

$$\int_0^\infty e^{-\alpha t} P^*(\tau_b \leq t) dt = \frac{1}{\alpha} e^{-bg(\alpha)}, \text{ for } \alpha > 0,$$

where $\tau_b = \tau(b, X)$ and $g(\cdot)$ is the inverse function of $G(\cdot)$.

Proof. Integration by parts is applied to solve the formula, that is,

$$\begin{aligned} \int_0^\infty e^{-\alpha t} P^*(\tau_b \leq t) dt \\ = \frac{1}{\alpha} \int_0^\infty e^{-\alpha t} dP^*(\tau_b \leq t) = \frac{1}{\alpha} E^*(e^{-\alpha \tau_b}), \end{aligned}$$

in which, from the definition of expectation, we can get the second equation. Then use the result derived from Fuh and Luo (2007) as follows:

$$E^*(e^{-\alpha \tau_b}) = e^{-bg(\alpha)}, \text{ for } \alpha > 0.$$

Since $G(\alpha)$ is an one-to-one function when $\alpha > 0$, $g(\alpha)$, the inverse function of $G(\alpha)$, exists and is also an one-to-one function, and therefore we complete the proofs. $\square$

3.2 Some known results

We have already known that the distribution of the continuous first passage time. The relation, however, between the distributions of $\tau(x, X)$ and $\tau'(x, X)\Delta t$ is more important for us. From Fuh and Luo (2007) we have something about the relations. The result is explained as follows and the proofs of the proposition in detail are referred to Fuh and Luo (2007).

Proposition 3.2. *We have*

$$\tau'(b, X)\Delta t - \tau(b + \beta C\sqrt{\Delta t}, X) \xrightarrow{d} 0,$$

where “ $\xrightarrow{d}$ ” means converging in distribution, and moreover

$$P^*(\tau'(b, X)\Delta t \leq T) = P^*(\tau(b + \beta C\sqrt{\Delta t}, X) \leq T) + o\left(\frac{1}{\sqrt{m}}\right).$$

In the proposition b is a fixed constant. However, we have to extend the results to a flexible number x which variates from a to ∞ for the proof of the main theorem. Since in this paper, the stopping level of first passage time is not a fixed constant. That is, $\tau(x, X)$ and $\tau'(x, X)$ will change according to level x . And in equations (3) and (5), we can also see that the two equations contain integrals with respect to the random variable x which is variable from a to ∞ . Therefore, We must discuss the uniform convergence of distributions of the first passage times when b is a variable number and the following lemma explains the properties.

Lemma 3.3. *Suppose that x is a flexible real number and $I = [a, \infty]$, where $a = \log(\frac{S_+}{s})$. Then we have that as $m \rightarrow \infty$*

$$P^*(\tau'(x, X)\Delta t \leq T) - P^*(\tau(x + \beta C\sqrt{\Delta t}, X) \leq T) \rightarrow 0$$

uniformly for $x \in I$. That is,

$$P^*(\tau'(x, X)\Delta t \leq T) = P^*(\tau(x + \beta C\sqrt{\Delta t}, X) \leq T) + o(\frac{1}{\sqrt{m}})$$

holds for all $x \in I$.

Here, x can be regarded as the *moving stopping level*. The state that the stopping level x approximates to infinity is just like that the drift of process X approximates to zero. The idea is referred to Siegmund (1985), chapter VIII and X. This is our heuristic argument and we only need a regular proof.

3.3 Proof of Theorem 2.1 for the LBP case

We can now prove Theorem 2.1. We only consider the case of a put since a call works similarly. Starting from the discrete price from (5) we have

$$\begin{aligned}
V_m(S_+) &= e^{-r(m-k)\Delta t} S_+ - S_k + e^{-r(m-k)\Delta t} \int_a^\infty s e^x P^*(\tau'(x, X) \leq m) dx \\
&= e^{-rT} S_+ - S_t + e^{-rT} \int_a^\infty s e^x P^*(\tau(x + \beta C \sqrt{\Delta t}, X) \leq T) dx + o\left(\frac{1}{\sqrt{m}}\right) \\
&= e^{-rT} S_+ - S_t + e^{-rT} \int_{a'}^\infty s e^{y - \beta C \sqrt{\Delta t}} P^*(\tau(y, X) \leq T) dy + o\left(\frac{1}{\sqrt{m}}\right) \\
&= e^{-\beta C \sqrt{\Delta t}} [e^{\beta C \sqrt{\Delta t}} e^{-rT} S_+ + e^{-rT} \int_{a'}^\infty s e^y P^*(\tau(y, X) \leq T) dy] - S_t + o\left(\frac{1}{\sqrt{m}}\right) \\
&= e^{-\beta C \sqrt{\Delta t}} V(S_+ e^{\beta C \sqrt{\Delta t}}) + (e^{-\beta C \sqrt{\Delta t}} - 1) S_t + o\left(\frac{1}{\sqrt{m}}\right),
\end{aligned}$$

in which the second equation - the key step - is just the results of Lemma 3.2, and a change of variables, $y = x + \beta C \sqrt{\Delta t}$, yields the third equation where $a' = \log\left(\frac{S_+}{s}\right) + \beta C \sqrt{\Delta t}$. Moreover, the last equation completes the proof of theorem 2.1 for LBP case. Note that T is used to represent the time to maturity ($T - t$) in order to simplify the notations.

The proof of Theorem 2.2 follows essentially the same steps. Alternatively, Theorem 2.2 can be derived from Theorem 2.1 by using the relation in (8).

4 Numerical Results

In this section, we develop numerical procedures which can compute the prices of the options considered in section 2. We begin with continuous lookback options, then consider discrete-version lookback options, and finally examine the accuracy of theorem 2.1 by correcting the continuous lookback options. Here, we merely consider the floating strike case. Note that any numerical method for pricing floating strike lookback options can be used to price fixed strike case.

4.1 Laplace transforms methods for continuous case

Laplace transforms have been widely used in valuing financial derivatives. This paper aims at using one-dimensional Laplace transforms to price the continuous lookback options. Because the calculation for the lookback call option follows just by symmetry, we will only provide the result for the lookback put option (LBP), whose price at an arbitrary time $t \in [0, T]$ is given by

$$\begin{aligned} LBP(T) &= e^{-r(T-t)} \mathbf{E}^*[\max_{t \in [0, T]} S_t - S(T) | \mathcal{F}_t] \\ &= e^{-r(T-t)} S_+ - S_t + e^{-r(T-t)} \int_a^\infty s e^x P^*(\tau_x \leq T) dx, \end{aligned} \quad (12)$$

where $S_+ \geq \max[S(t), s]$ is the predetermined max, $a = \log \frac{S_+}{s}$ and $\tau_x = \tau(x, X)$.

Following the same argument in Kou and Wang (2004), we can obtain the one-dimensional Laplace transform of (12) under the spectrally negative jump processes.

That is when $0 < \delta < 1$.

Proposition 4.1. *For any given α such that $\alpha + r > 0$, the Laplace transform with respect to T of the LBP option is given by*

$$\begin{aligned} & \int_0^\infty e^{-\alpha T} LBP(T) dT \\ &= \frac{S_+ e^{rt}}{\alpha + r} - \frac{S_t}{\alpha} + \frac{se^{rt+a[1-g(\alpha+r)]}}{(\alpha + r)[g(\alpha + r) - 1]}, \end{aligned}$$

here $g(\cdot)$ is the inverse function of $G(\cdot)$.

Proof. It follows from (12) and the Fubini theorem that

$$\begin{aligned} f(\alpha) &= \int_0^\infty e^{-\alpha T} \left[e^{-r(T-t)} S_+ - S_t + e^{-r(T-t)} \int_a^\infty se^x P^*(\tau_x \leq T) \right] dT \\ &= \int_0^\infty e^{-(\alpha+r)T+rt} S_+ - e^{-\alpha T} S_t dT \\ &\quad + \int_0^\infty \int_a^\infty e^{-(\alpha+r)T+rt} se^x P^*(\tau_x \leq T) dx dT \\ &= \frac{S_+ e^{rt}}{\alpha + r} - \frac{S_t}{\alpha} + se^{rt} \int_a^\infty e^x \int_0^\infty e^{-(\alpha+r)T} P^*(\tau_x \leq T) dT dx. \end{aligned} \tag{13}$$

By using Lemma 3.1, the last term becomes

$$\begin{aligned} & \frac{se^{rt}}{\alpha + r} \int_a^\infty e^x \cdot e^{-xg(\alpha+r)} dx \\ &= \frac{se^{rt}}{\alpha + r} \int_a^\infty e^{x[1-g(\alpha+r)]} dx \\ &= \frac{se^{rt+a[1-g(\alpha+r)]}}{(\alpha + r)[g(\alpha + r) - 1]}. \end{aligned}$$

This completes the proof. □

Since the solutions for lookback options are given in terms of Laplace transforms, numerical inversion of Laplace transforms is required. To do this, we shall use the Gaver-Stehfest algorithm⁸. Given the Laplace transform function $f(\alpha)$ of the $LBP(T)$, the algorithm generates a sequence $f_n(T)$ such that $f_n(T) \rightarrow LBP(T)$ as $n \rightarrow \infty$. Here we show the inverse formulae as follows:

$$LBP(T) \cong \left[\frac{\log 2}{T} \right] \sum_{j=1}^n d(j, n) f(j \log[2]/T),$$

where n is an even integer, $M = n/2$, m is the integer part of $(j+1)/2$ and

$$d(j, n) = (-1)^{j+M} \sum_{k=m}^{\min[j, M]} \frac{k^M (2k)!}{(M-k)! k! (k-1)! (j-k)! (2k-j)!}.$$

The above formulae give the Gaver-Stehfest numerical inversion method we used. The algorithm is simple and converges very fast; as we will see, it typically converges nicely even for J between 4 and 10. Kou and Wang (2003) reported the details of implementation.

As a numerical illustration, we calculate the continuous lookback put option in Table 1. For convenience, we just use Mathematica 4.0 to run the algorithm.

⁸For an argument in favor of using this algorithm, we refer to Kou and Wang (2003).

Table 1: Continuously monitored LBP option, varing λ and J .

	$\lambda = 1$		$\lambda = 2$		$\lambda = 4$	
n	$V(S_+)$	Approx.(1)	$V(S_+)$	Approx.(1)	$V(S_+)$	Approx.(1)
4	13.3177	12.2128	14.6740	13.4831	17.2905	15.9330
6	12.2639	11.1722	13.5771	12.3999	16.1151	14.7726
8	12.2804	11.1886	13.5926	12.4153	16.1287	14.7863
10	12.2827	11.1907	13.5951	12.4176	16.1316	14.7890

Option parameters: $s = 95$, $S_+ = 110$, $r = 0.1$, $\sigma = 0.3$, $\delta = 0.9$, $T = 1$ and $t = 0.8$. Assume 250 trading days per year, $m = 250$ monitoring points roughly corresponds to daily monitoring of the extreme. The Approx.(1) is obtained by Theorem 2.1.

Table 1 shows that the algorithm converges when $J = 10$. Besides, the bigger λ is, the larger the prices are. It can be thought that when the frequency of jumps within a year increases, terminal price of the underlying asset would decrease. Furthermore, the spread of maximal and terminal price would widen which cause the floating strike lookback put options become expensive.

If we suppose that the Theorem 2.1 holds, that is the Approx.(1) is the true price of discrete lookback put option, then it can be observed that the difference between continuous LBP and discrete one is obviously significant. This is the reason why we can not use the formulae of continuous LBP options to price discrete cases.

4.2 Pricing discrete LBP and numerical results

As for the computation of the price of the discrete lookback options, we simply use Monte Carlo simulation method with 10^5 replications. The simulation prices have an average accuracy of about 0.001 for the cases we will consider later. And we will use these values as the benchmark (true value).

Table 2 compares the approximation price values with those of discrete lookback options for different numbers of monitoring points m . The options considered are floating strike put options and the option parameters we used here are: $s = 95$, $S_+ = 110$, $r = 0.1$, $\sigma = 0.3$, $\delta = 0.9$, $\lambda = 1$, $T = 1(\text{year})$ and $t = 0.8(\text{year})$. Assuming 250 trading days per year. In column three, the corrected continuous price is the approximation of Theorem 2.1 for pricing discrete LBP options with a predetermined maximum. And In column four, the absolute error is calculated by Corrected continuous price $-V_m(S_+)$ while in the last column, the relative error is given by $AE/V_m(S_+)$. And we can see that the resulting approximation is good, especially for larger m . Furthermore, we can see that the continuous LBP $V(S_+)$ with $S_+ = 110$ is 13.3177 which is calculated by Laplace transform method ;the difference between $V(S_+)$ and the true value is large. Hence, using the approximation formulae in Theorem 2.1 to price discrete lookback options is more accurate.

Table 2: Lookback Put Option Price Results, $S_+ = 110$, varying m

m	Discrete lookback $V_m(S_+)$	Corrected continuous lookback(1)	Absolute error AE	Relative error RE(%)
10	9.9874	8.2146	-1.7728	-17.75
20	10.5995	9.6022	-0.9973	-9.41
40	11.2436	10.6359	-0.6077	-5.40
80	11.8076	11.3937	-0.4139	-3.51
160	12.0339	11.9433	-0.0906	-0.75
250	12.2346	12.2128	-0.0218	-0.18

Option parameters: $s = 95$, $S_+ = 110$, $r = 0.1$, $\sigma = 0.3$, $\delta = 0.9$, $\lambda = 1$, $T = 1$ and $t = 0.8$. Assume 250 trading days per year. The continuous LBP $V(S_+)$ is 13.3177 and the price (1) is the approximative value in Theorem 2.1

In order to assess the effectiveness of the approximation in Theorem 2.1, we priced discrete lookback put options with different monitoring frequencies m and three values for the predetermined maximum S_+ , while the other option parameters are the same with those in Table 2. The results are presented in Table 3 and show that the overall approximation is getting better as m increasing which is consistent with our intuition. In addition, smaller errors are obtained when the predetermined max S_+ is further from the current asset price S_t . In Table 4 we compare with different values of δ which affect the variance of the Poisson process part. We see that when the jump size, δ , is getting lower (farther from 1), the performance of the approximation is getting worse. In summary, our suggesting approximation is accurate especially for high frequency of monitoring, far level of the predetermined maximum from the current asset price S_t , and closer value of the jump size to 1.

Table 3: Lookback Put Option Price Results, varing m and S_+

		Discrete	Corrected	Absolute	Relative
		lookback	continuous	error	error
S_+	m	$V_m(S_+)$	lookback(1)	AE	RE(%)
105	10	6.2338	4.3655	1.8683	29.97
	40	8.7566	7.2850	1.4716	16.80
	80	9.8155	8.3611	1.4544	14.82
	160	9.5242	8.0714	1.4528	15.25
	250	10.1299	9.4194	0.7105	7.01
110	10	9.9874	8.2146	1.7728	17.75
	40	11.2436	10.6359	0.6077	5.40
	80	11.5467	11.3937	0.153	1.33
	160	11.9873	11.9433	0.044	0.37
	250	12.2346	12.2128	0.0218	0.18
115	10	14.1987	13.1950	1.0037	7.00
	40	14.8044	14.6757	0.1287	0.87
	80	15.0148	15.2099	0.1951	1.29
	160	15.5154	15.6172	0.1018	0.66
	250	15.8215	15.8229	0.0014	0.00

Option parameters: $s = 95$, $r = 0.1$, $\sigma = 0.3$,
 $\delta = 0.9$, $\lambda = 1$, $T = 1$ and $t = 0.2$. Assume 250
trading days per year.

Table 4: Lookback Put Option Price Results, varing m and δ

		Discrete	Corrected	Absolute	Relative
		lookback	continuous	error	error
δ	m	$V_m(S_+)$	lookback(1)	AE	RE(%)
0.3	10	11.9853	6.1263	5.8590	48.88
	40	16.1883	10.8796	5.3087	32.79
	80	17.4901	13.7712	3.7189	21.26
	160	18.5497	15.9272	2.6225	14.14
	250	18.9415	17.0052	1.9463	10.27
0.6	10	10.5416	7.0381	3.5035	33.23
	40	13.2894	11.381	1.9084	14.36
	80	14.3262	13.0857	1.2405	8.66
	160	14.9641	14.3328	0.6313	4.22
	250	15.1773	14.9475	0.2298	1.51
0.9	10	9.9874	8.2146	1.7728	17.75
	40	11.2436	10.6359	0.6077	5.40
	80	11.5467	11.3937	0.153	1.33
	160	11.9873	11.9433	0.044	0.37
	250	12.2346	12.2128	0.0218	0.18

Option parameters: $s = 95$, $r = 0.1$, $\sigma = 0.3$,
 $S_+ = 110$, $\lambda = 1$, $T = 1$ and $t = 0.2$. Assume 250
trading days per year.

5 Conclusion Remarks and Further Research

we have addressed the problem of pricing Lookback options depending on extremal values of the underlying asset when the extremal values are monitored over a discrete set of dates rather than in continuous time. As in the case under Black-Scholes model, there is also a continuity correction for discrete lookback options under constant jump diffusion model. The correction terms allow the discrete versions to be quite accurately using formulae for the continuous versions. We can see that the correction terms shift mainly the predetermined maximal value of the underlying asset with two factors: C and β . The former is different from that in Black-Scholes model which is just σ . Note that C is larger than σ and this represents the higher variance of a jump diffusion model which usually has heavy tails. The second term is the same due to Siegmund's corrected diffusion approximation. Finally, we applied a fast and convenient numerical inverse Laplace transforms methods to evaluate the approximations and to price the options with high accuracy. We can thus get the discrete option price more quickly.

As for further works, we have to deal with the uniform convergence issue of the distributions of stopping times first, in which we need a proof of Lemma 3.3 to complete the whole theory. Besides this, we also would like to see if we can extend similar corrected approximations to more general jump diffusion models, for instance, double exponential jump diffusion model which has analytical solutions for

many path-dependent options. On the other hand, we would like to go a step further to analyze what if the condition becomes $\delta > 1$ for LBP options while $0 < \delta < 1$ for LBC cases in Theorem 2.1. The difficulty of the conditions is that there is an overshoot phenomenon even in the continuously monitoring case and it is not easy to make the difference between the discrete and continuous overshoot effects. That is, it is hard to tell apart the overshoot due to jump effect from due to discretization effect. This is an interesting open problem and deserves further study.

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